

ADC Fall'25 Project Proposal

Project title: RF Protocol Classification and Identification Using Software Defined Radio (SDR)

Objective and Motivation:

The objective of this project is to design and implement a system capable of automatically identifying and classifying radio frequency (RF) signals based on their underlying communication protocols using a Software Defined Radio (SDR) and machine learning techniques.

In modern wireless environments, countless devices operate simultaneously using different communication standards such as Wi-Fi, Bluetooth, GSM, and FM radio. Identifying and categorizing these signals is crucial for applications in spectrum monitoring, interference detection, and intelligent radio systems.

This project is motivated by the growing need for intelligent spectrum awareness, where systems can autonomously understand their RF surroundings without human interpretation or protocol-specific decoders. The outcome will contribute toward the foundation of AI-driven RF analysis systems — useful for both academic research and real-world applications such as cognitive radios, IoT monitoring, and electronic reconnaissance.

Methodology / Approach:

The project will follow a modular approach divided into distinct stages:

a. Signal Acquisition

- Use the HackRF One SDR to capture I/Q samples from different RF bands (FM, Wi-Fi, Bluetooth, GSM, etc.).
- Record signals under varied conditions (distance, power, interference) to ensure data diversity.

b. Preprocessing

- Apply filtering, normalization, and windowing on raw I/Q data.
- Convert time-domain samples into spectrograms (time–frequency representations) using Short-Time Fourier Transform (STFT).

c. Feature Extraction

- Extract features such as:
 - Power Spectral Density (PSD)
 - Bandwidth and center frequency

- Spectrogram image data for machine learning models

d. Model Development

- Train a Convolutional Neural Network (CNN) on labeled spectrogram images for protocol classification.
- Alternatively, test classical ML models (Random Forest, SVM) for comparison.
- Use a supervised learning setup, where each signal segment is labeled according to its protocol.

e. Evaluation

- Validate model performance on unseen test data.
- Measure accuracy and detection latency.

Tools and equipment required:

I. Hardware:

I. HackRF One

II. Omnidirectional antenna

II. Software:

I. GNU Radio / SDR# / GQRX

II. Python (NumPy, SciPy, scikit-learn, PyTorch/TensorFlow, Matplotlib)

Dataset plan:

I. Data Collection

- Signals will be recorded using an RTL-SDR receiver connected to a laptop running GNU Radio or Python-based capture scripts.
- Each protocol will be recorded at its respective operating frequency bands:
 - FM Radio: 88–108 MHz
 - Bluetooth/Wi-Fi: 2.4 GHz
 - GSM: 900 MHz
 - LoRa: 433 MHz

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- Multiple 10–30 second recordings will be made for each protocol under varying SNR and distance conditions.

II. Data Labeling and Preprocessing

- Each recording will be labeled with its known protocol.
- Raw I/Q samples will be segmented into 1-second windows.
- Windows will be transformed into spectrograms using STFT, creating a labeled dataset suitable for CNN input.

III. Data Storage

- All data will be stored in a structured format (. npy or . csv), with metadata including frequency, bandwidth, timestamp, and environment notes.

Expected Output / Performance Metric:

I. Expected Output

A working prototype of a RF signal classifier capable of identifying protocols from SDR input.

- A dashboard (CLI or GUI) displaying:
 - Detected protocol name
 - Frequency band
 - Confidence score

II. Performance Metrics

- Classification Accuracy: $\geq 85\%$ across all test protocols.
- Scalability: Ability to add new protocols with retraining.

Timeline and Work Distribution:

I. Timeline: The project is aimed to be completed by the end of semester.

II. Work distribution: Both members would be putting in equal time and working together over the entire project.